

Physics 1: Units, Measurement & Motion

A first-time explanation — no prior knowledge assumed

How to read this chapter

- 15 short lessons. Each one explains the idea from scratch, shows a picture, works an example step by step, and ends with a **Quick check**.
- Answer each Quick check **out loud before moving on**. If you can't, re-read only that lesson, not the whole chapter.
- First reading: about 50 minutes. After that, revise with the 1-page summary and the practice set at the end.
- **Part A** (Lessons 1–8): units & measurement. **Part B** (Lessons 9–15): motion.

#	Lesson	#	Lesson
1	What 'measuring' means	9	Rest and motion are relative
2	Base and derived quantities	10	Distance vs displacement
3	Named units — what they really mean	11	Speed, velocity, average speed
4	Very big and very small units	12	Acceleration
5	Temperature scales	13	Reading motion graphs
6	Measuring instruments	14	Equations of motion
7	Scalars and vectors	15	Free fall and throwing upward
8	Dimensions	–	Summary · Practice set · Answers

PART A — Units & Measurement

Lesson 1 · What does 'measuring' mean? (about 2 min)

To **measure** something means to **compare it with a fixed, agreed amount**, called a **unit**. When you say 'this table is 2 metres long', you are saying the table is **2 times** as long as one metre.

So every measurement has **two parts**: a **number** and a **unit**. '2' on its own means nothing — 2 metres? 2 feet? 2 hand-spans? Only '2 m' is a complete measurement.

Why we need a standard unit. Long ago people measured with body parts: hand-span, foot, cubit (elbow to fingertip). The problem: your hand-span and mine are different sizes. The same table might be '8 spans' for me and '9 spans' for you. So the world agreed on one common system in **1960**: the **SI system** (from the French *Système International*, meaning **International System of Units**). The metre, kilogram and second all belong to it.

Quick check 1 — say the answers out loud before moving on (answers on the last page)

- a) What are the two parts of every measurement? b) Why can't a hand-span be a standard unit?
- c) What is the full form of SI?

Lesson 2 · Base (fundamental) and derived quantities (about 4 min)

Think of the English alphabet. With just 26 letters, you can write every word. Physics works the same way: there are **7 base quantities** (the 'letters'), and every other quantity (the 'words') is built from them.

- A **base (fundamental) quantity** cannot be broken into simpler quantities — e.g. length, mass, time.
- A **derived quantity** is made by multiplying or dividing base quantities — e.g. speed, area, force.

See it happen: a room is 4 m long and 3 m wide. $\text{Area} = 4 \text{ m} \times 3 \text{ m} = 12 \text{ m} \times \text{m} = 12 \text{ m}^2$. A runner covers 100 m in 10 s. $\text{Speed} = 100 \text{ m} \div 10 \text{ s} = 10 \text{ m} \div \text{s} = 10 \text{ m/s}$. You didn't need to memorise 'unit of speed is m/s' — the unit **comes out of the formula**. This trick works for every derived unit:

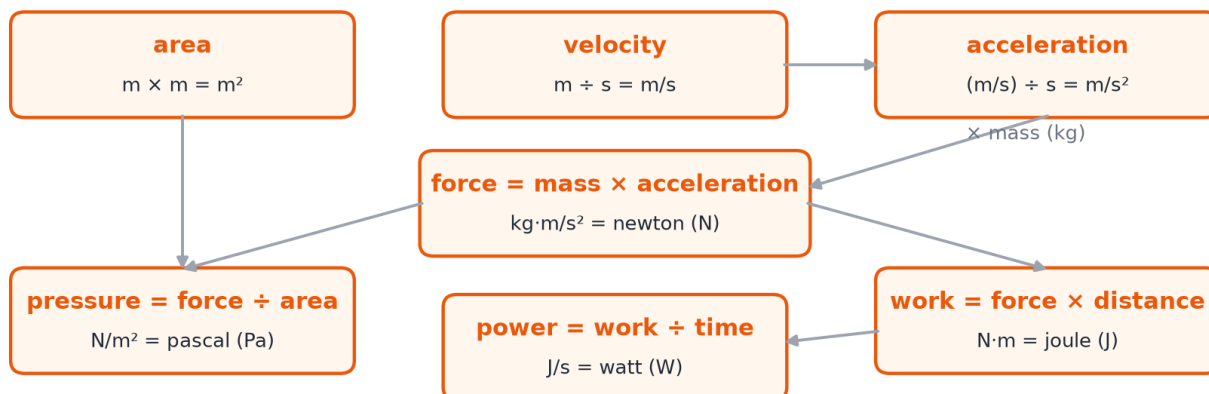
BASE units (the 'letters')

metre (m)
length

second (s)
time

kilogram (kg)
mass

DERIVED units (the 'words') — each is built from the base units



The 7 base units, and what each one measures in plain words:

m	kg	s	A	K	mol	cd
metre	kilogram	second	ampere	kelvin	mole	candela
Length	Mass	Time	Electric current	Temperature	Amount of substance	Luminous intensity

Base quantity	SI unit	Plain meaning
Length	metre (m)	How long or far
Mass	kilogram (kg)	How much matter is in an object (not the same as weight)
Time	second (s)	How long something takes
Electric current	ampere (A)	How much electric charge flows per second
Temperature	kelvin (K)	How hot or cold
Amount of substance	mole (mol)	A counting unit for tiny particles — like 'dozen' means 12, one mole means 6.022×10^{23} particles
Luminous intensity	candela (cd)	How bright a light source is in a given direction

Remember

- **kg** is the only base unit with a prefix ('kilo') already in its name.
- Temperature's SI unit is **kelvin (K)**, not °C. Write 'K', never '°K'.
- **Candela** is the base unit. Lumen and lux are derived units of light — common wrong options.

Quick check 2 — say the answers out loud before moving on (answers on the last page)

a) Is speed a base or a derived quantity? b) What is the SI unit of volume? (Hint: length × length × length) c) What is the SI base unit of temperature?

Lesson 3 · Named derived units — what they actually mean (about 5 min)

Some derived units are named after scientists. **Rule:** the **symbol starts with a capital** (N, J, W, Pa, Hz), but the **name is written in small letters** (newton, joule, watt). Here's what the five most important ones actually feel like:

Unit	Measures	What it means	Real-life feel
newton (N)	Force	The force that makes a 1 kg mass speed up by 1 m/s every second	Holding a 100 g apple needs about 1 N
joule (J)	Work / energy	The work done when a 1 N force moves something 1 m	Lifting that apple 1 m up $\approx$ 1 J
watt (W)	Power	1 joule of energy used per second	A 100 W bulb uses 100 J every second. Power = how fast energy is used
pascal (Pa)	Pressure	A 1 N force spread over 1 m ² — a very small pressure	Air pressure around you $\approx$ 1,00,000 Pa
hertz (Hz)	Frequency	1 vibration or cycle per second	Indian mains electricity is 50 Hz

Pressure = force $\div$ area, so the **same force on a smaller area gives more pressure**.

That's why a sharp knife cuts better than a blunt one, and why a camel's wide feet don't sink into sand.

The electrical units below come back on Day 7. For now, just learn to match each quantity to its unit:

Quantity	Unit (symbol)	Built from
Electric charge	coulomb (C)	ampere $\times$ second
Potential difference (voltage)	volt (V)	joule $\div$ coulomb
Resistance	ohm (Ω)	volt $\div$ ampere
Conductance	siemens (S) — older name mho	1 $\div$ ohm
Capacitance	farad (F)	coulomb $\div$ volt
Magnetic flux	weber (Wb)	volt $\times$ second
Magnetic field strength	tesla (T)	weber $\div$ m ²
Inductance	henry (H)	weber $\div$ ampere
Radioactivity	becquerel (Bq) — older unit curie	1 decay per second
Power of a lens	diopetre (D)	1 $\div$ focal length in metres
Illuminance	lux (lx)	lumen $\div$ m ²

Worked example — power

A motor does 1200 J of work in 1 minute. What is its power?

Step 1: convert the time to seconds: 1 min = 60 s. Step 2: power = work $\div$ time = 1200 $\div$ 60 = **20 W**.

Quick check 3 — say the answers out loud before moving on (answers on the last page)

a) A machine does 600 J of work in 2 minutes. What is its power? b) Which gives more pressure: 100 N on 2 m², or 100 N on 0.5 m²? c) What is the symbol for the unit named after Pascal?

Lesson 4 • Very big and very small units (about 5 min)

Writing the distance to a star in metres gives absurdly long numbers, and so does writing the size of an atom. So scientists use **special units** for very big and very small things. Here they are, smallest to largest:



Light year — the biggest trap in this chapter. The word 'year' makes people think it's a unit of time. It isn't. A light year is the **distance light travels in one year**. Let's calculate it:

Worked example — how big is a light year?

Light travels 3×10^8 m every second.

Seconds in a year = $365 \times 24 \times 60 \times 60 \approx 3.15 \times 10^7$ s.

Distance = speed $\times$ time = $3 \times 10^8 \times 3.15 \times 10^7 \approx \mathbf{9.46 \times 10^{15} \text{ m}}$. That's a **distance**.

Astronomical unit (AU) = the average Sun–Earth distance, about 1.5×10^{11} m. **Parsec** = 3.26 light years — the largest of the three. Order: **AU < light year < parsec**.

Kilowatt-hour (kWh) — the 'unit' on your electricity bill. A kilowatt is **power** (how fast energy is used). Multiply power by time and you get **energy**. So kWh is a unit of **energy**, not power.

Worked example — electricity bill

1 kWh = $1000 \text{ J/s} \times 3600 \text{ s} = \mathbf{3.6 \times 10^6 \text{ J}}$.

A 1.5 kW AC runs for 4 hours: energy = $1.5 \times 4 = \mathbf{6 \text{ kWh} = 6 \text{ units}}$.

A 100 W bulb runs for 10 hours: $0.1 \text{ kW} \times 10 \text{ h} = \mathbf{1 \text{ unit}}$.

Other units to recognise:

Unit	What it measures	Value / note
Horsepower (HP)	Power	1 HP = 746 W. James Watt compared his steam engines with horses
Nautical mile	Distance at sea and in the air	1852 m
Knot	Speed of ships and aircraft	1 nautical mile per hour — it's a speed, not a distance
Fathom	Depth of water	6 feet $\approx$ 1.83 m
Calorie (cal)	Heat energy	1 cal $\approx$ 4.186 J (often 4.2 J). A food 'Calorie' = 1 kcal = 1000 cal
Electron-volt (eV)	Energy of tiny particles	1 eV = 1.6×10^{-19} J
Atmosphere (atm)	Pressure	1 atm = 1.013×10^5 Pa = 76 cm (760 mm) of mercury
bar / torr	Pressure	1 bar = 10^5 Pa · 1 torr = 1 mm of mercury
Decibel (dB)	Loudness (sound intensity level)	Comes back on Day 5

Exam trap

Light year → **distance**. Knot → **speed**. kWh → **energy**. HP → **power**. Fathom → **depth**.
When the question asks 'which is NOT a unit of distance?', look for the speed or time unit hidden among the options.

Quick check 4 — say the answers out loud before moving on (answers on the last page)

a) A 2 kW heater runs for 3 hours. How many units of electricity does it use? b) Arrange from smallest to largest: parsec, AU, light year. c) A ship moves at 20 knots. Is a knot a unit of speed or of distance?

Lesson 5 · Temperature scales (about 5 min)

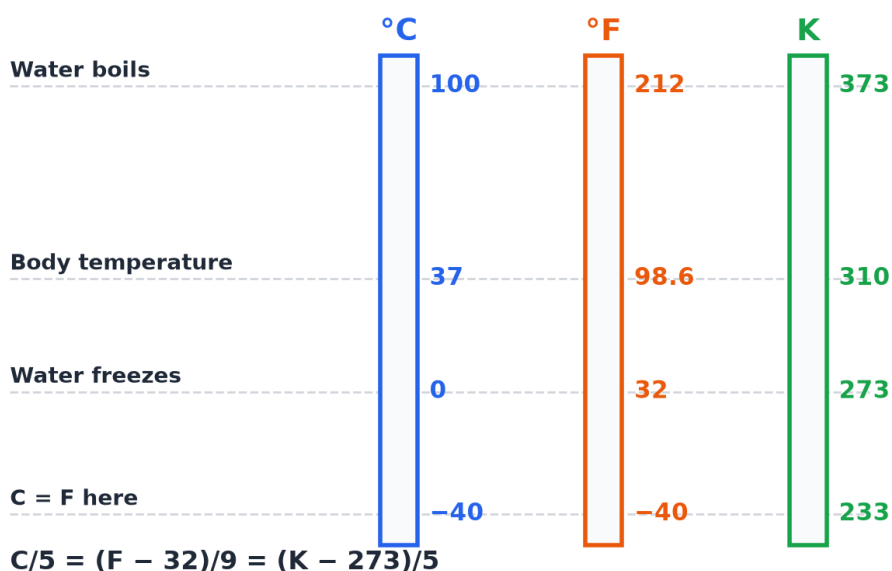
Temperature tells you how hot or cold something is. Heat always flows from the hotter object to the colder one. (Trap: **heat** is energy, measured in joules; **temperature** is measured in kelvin or °C.)

Why the three scales give different numbers. Think of measuring your height in feet and in centimetres: the height is the same, only the numbers differ. Each temperature scale picks two fixed points — ice melting and water boiling — and gives them different numbers:

Scale	Ice melts	Water boils	Number of steps between them
Celsius (°C)	0	100	100
Fahrenheit (°F)	32	212	180
Kelvin (K)	273	373	100 (the same step size as Celsius)

So 100 Celsius steps cover the same range as 180 Fahrenheit steps: **1 °C step = 1.8 °F steps**. Writing 'how far up from ice ÷ total steps' for each scale gives:

$$C/100 = (F - 32)/180 = (K - 273)/100 \rightarrow \text{divide everything by 20} \rightarrow C/5 = (F - 32)/9 = (K - 273)/5$$



Worked example — four conversions

(1) $25\text{ }^{\circ}\text{C} \rightarrow ^{\circ}\text{F}$: $25/5 = 5 \rightarrow (F - 32)/9 = 5 \rightarrow F - 32 = 45 \rightarrow F = 77\text{ }^{\circ}\text{F}$. In kelvin: $25 + 273 = 298\text{ K}$.

(2) $98.6\text{ }^{\circ}\text{F} \rightarrow ^{\circ}\text{C}$: $F - 32 = 66.6 \rightarrow 66.6/9 = 7.4 \rightarrow C = 7.4 \times 5 = 37\text{ }^{\circ}\text{C}$ (normal body temperature).

(3) **Where are C and F equal?** Put $C = F = x$: $x/5 = (x - 32)/9 \rightarrow 9x = 5x - 160 \rightarrow x = -40$.

(4) **Absolute zero** = $0\text{ K} = -273\text{ }^{\circ}\text{C}$ (more exactly $-273.15\text{ }^{\circ}\text{C}$), the coldest possible temperature. Kelvin has **no negative values**.

Change vs value: if the temperature **rises** by $10\text{ }^{\circ}\text{C}$, it rises by 10 K and by $18\text{ }^{\circ}\text{F}$. For a change, you multiply by 1.8 and do **not** add 32.

Quick check 5 — say the answers out loud before moving on (answers on the last page)

a) Convert $50\text{ }^{\circ}\text{C}$ to $^{\circ}\text{F}$. b) Convert 300 K to $^{\circ}\text{C}$. c) The temperature rises by $20\text{ }^{\circ}\text{C}$. By how many $^{\circ}\text{F}$ does it rise?

Lesson 6 · Measuring instruments — learn the word roots (about 5 min)

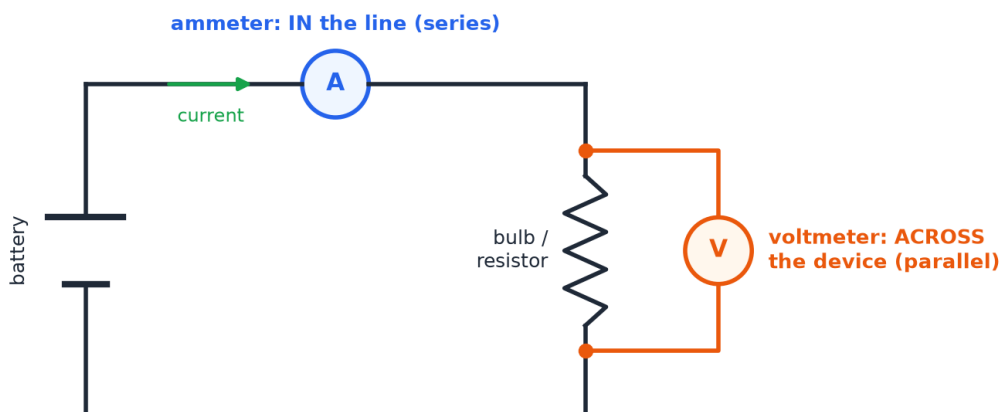
Don't memorise instruments as a random list. Most of their names are made from a **Greek or Latin root** plus '-meter' (measurer). Learn the root and you can work out the instrument:

Root	Means	Instrument	Measures
baro-	weight / pressure	Barometer	Atmospheric pressure
hygro-	moisture	Hygrometer	Relative humidity of the air
hydro-	water	Hydrometer	Relative density of liquids
lacto-	milk	Lactometer	Purity of milk (its density)
mano-	thin / rare (gas)	Manometer	Pressure of gases
sphygmo-	pulse	Sphygmomanometer	Blood pressure
seismo-	shaking	Seismograph	Earthquake waves
anemo-	wind	Anemometer	Wind speed
alti-	height	Altimeter	Altitude (height above sea level)
pyro-	fire	Pyrometer	Very high temperatures (furnaces)
odo-	path / way	Odometer	Distance travelled by a vehicle
tacho-	speed	Tachometer	Rotation speed of an engine (rpm)

Exam trap

Hydrometer (liquid density) vs **hygrometer** (humidity) — one letter apart, and a favourite exam option pair.

Ammeter and voltmeter — why they are connected differently



- An **ammeter** counts the current flowing **through** a wire, so it must sit **in the path** of the current (**series**). It has a **very low resistance**, so it doesn't block the current it's measuring.
- A **voltmeter** compares the 'electrical push' at **two points**, so it's connected **across** the device (**parallel**). It has a **very high resistance**, so almost no current is diverted through it.
- A **galvanometer** only **detects** small currents. It becomes an **ammeter** with a low resistance (a 'shunt') added in parallel, and a **voltmeter** with a high resistance added in series.

Least count — the smallest reading an instrument can make

Instrument	How the least count is worked out	Least count	Typical use
Metre scale (ruler)	Smallest marking	1 mm	Length of a book
Vernier calliper	1 main-scale division – 1 vernier division = 1 mm – 0.9 mm	0.1 mm (0.01 cm)	Diameter of a pipe or bob
Screw gauge	Pitch ÷ number of circular divisions, e.g. 0.5 mm ÷ 50	0.01 mm (0.001 cm)	Thickness of a wire or sheet

The smaller the least count, the more precise the instrument. So **screw gauge > vernier > ruler** for precision. A clinical thermometer reads **35 °C to 42 °C** (about 94–108 °F).

Quick check 6 — say the answers out loud before moving on (answers on the last page)

a) Which instrument measures humidity? b) How is a voltmeter connected in a circuit? c) Which is more precise: a vernier calliper or a screw gauge?

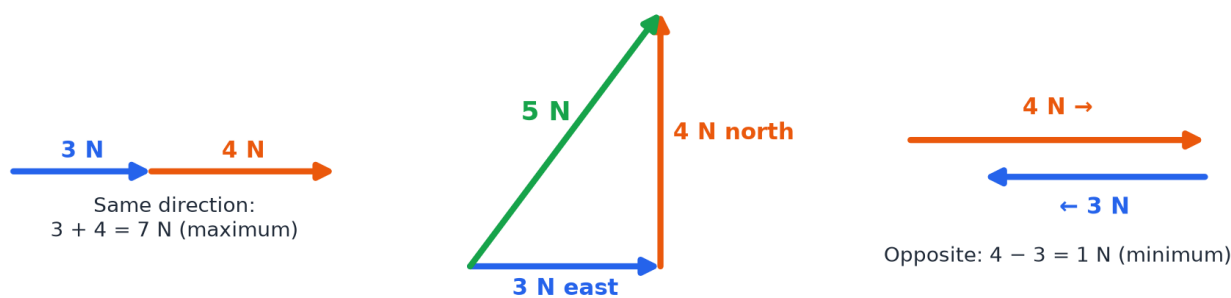
Lesson 7 · Scalars and vectors (about 5 min)

Your friend says: 'Walk 5 metres and you'll find the treasure.' Can you find it? **No** — you don't know **which way**. 'Walk 5 m **east**' is enough. Some quantities are only complete when they include a **direction**.

- **Scalar**: has only size (magnitude). Example: 5 kg, 30 °C, 10 s.
- **Vector**: has size **and** direction, **and** adds like arrows. Example: 5 m east, 10 N downward.

Why direction changes the answer. Scalars add simply: 3 kg + 4 kg is always 7 kg. But a 3 N force plus a 4 N force could give anything from 1 N to 7 N, depending on their directions:

Adding a 3 N and a 4 N force: the answer depends on DIRECTION (that's what makes force a vector)



At right angles: $\sqrt{(3^2 + 4^2)} = 5 \text{ N}$

Rule: two vectors A and B combine to give anything from **(A – B)** (opposite directions) to **(A + B)** (same direction). At right angles, the result is $\sqrt{(A^2 + B^2)}$.

Scalars (size only)	Vectors (size + direction)
Mass, distance, speed, time, work, energy, power, temperature, pressure, density, volume, charge, electric current	Displacement, velocity, acceleration, force, weight , momentum, impulse, torque, electric field, magnetic field

Exam trap

- **Electric current** flows in a direction, but it's a **scalar**. At a junction, $2 \text{ A} + 3 \text{ A} = 5 \text{ A}$ whatever angle the wires meet at — currents don't add like arrows.
- **Pressure** is a scalar: at a point in a fluid it pushes equally in all directions. Force, however, is a vector.
- **Mass** is a scalar, but **weight** is a force (it pulls downward), so weight is a vector.

Quick check 7 — say the answers out loud before moving on (answers on the last page)

a) Forces of 6 N and 8 N act at right angles. What is the resultant? b) What range of values can 6 N and 8 N combine to give? c) Is electric current a vector?

Lesson 8 • Dimensions (about 5 min)

The **dimension** of a quantity shows **which base quantities it's built from, and to what power**. We use **M** for mass, **L** for length and **T** for time, written in square brackets.

Method: write the formula, then replace each part with its dimension. Never memorise these — **derive them**:

Quantity	Formula	Substitute	Dimension
Velocity	distance ÷ time	$L \div T$	$[L T^{-1}]$
Acceleration	velocity ÷ time	$L T^{-1} \div T$	$[L T^{-2}]$
Force	mass × acceleration	$M \times L T^{-2}$	$[M L T^{-2}]$
Momentum	mass × velocity	$M \times L T^{-1}$	$[M L T^{-1}]$
Work / energy	force × distance	$M L T^{-2} \times L$	$[M L^2 T^{-2}]$
Torque	force × distance	$M L T^{-2} \times L$	$[M L^2 T^{-2}]$ — same as work!
Power	work ÷ time	$M L^2 T^{-2} \div T$	$[M L^2 T^{-3}]$
Pressure / stress	force ÷ area	$M L T^{-2} \div L^2$	$[M L^{-1} T^{-2}]$

Same-dimension pairs (a frequent question type) appear when two quantities are built the same way: **work-energy-torque** · **pressure-stress-Young's modulus** · **impulse-momentum** · **Planck's constant-angular momentum** [$M L^2 T^{-1}$].

Dimensionless quantities are a ratio of two things of the same kind, so the dimensions cancel: **strain** (change in length $\div$ length), **refractive index** (speed $\div$ speed), **angle** (arc $\div$ radius), **relative density** (density $\div$ density).

Quick check 8 — say the answers out loud before moving on (answers on the last page)

a) What is the dimension of momentum? b) Which has the same dimension as pressure: stress or force? c) Does refractive index have a dimension?

PART B — Motion

Lesson 9 · Rest and motion are relative (about 2 min)

An object is **in motion** if its **position changes with time** compared with a reference point. But the answer depends on **who is watching**:

For Sita, Ravi is at REST (distance between them doesn't change).

For the person on the platform, Ravi is in MOTION. Rest and motion depend on the observer.

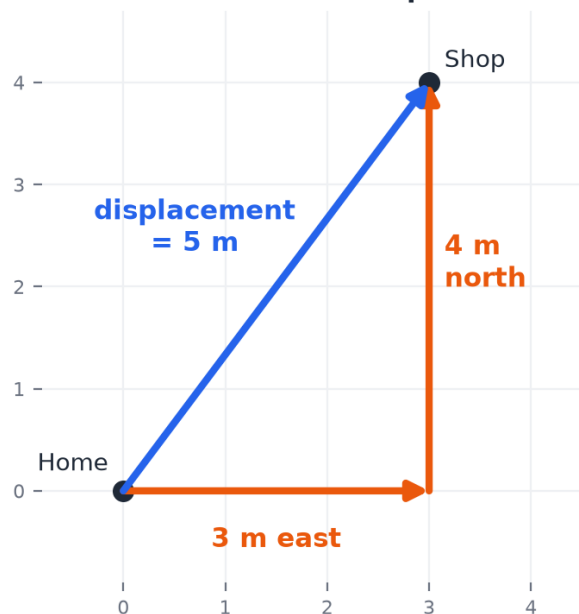


The same person can be at rest and in motion at the same time, depending on the observer. This is why we say **rest and motion are relative**. (Even a 'still' house is moving with the Earth as it spins and orbits the Sun.)

Lesson 10 · Distance vs displacement (about 4 min)

Imagine you walk from home 3 m east, then 4 m north, to reach a shop.

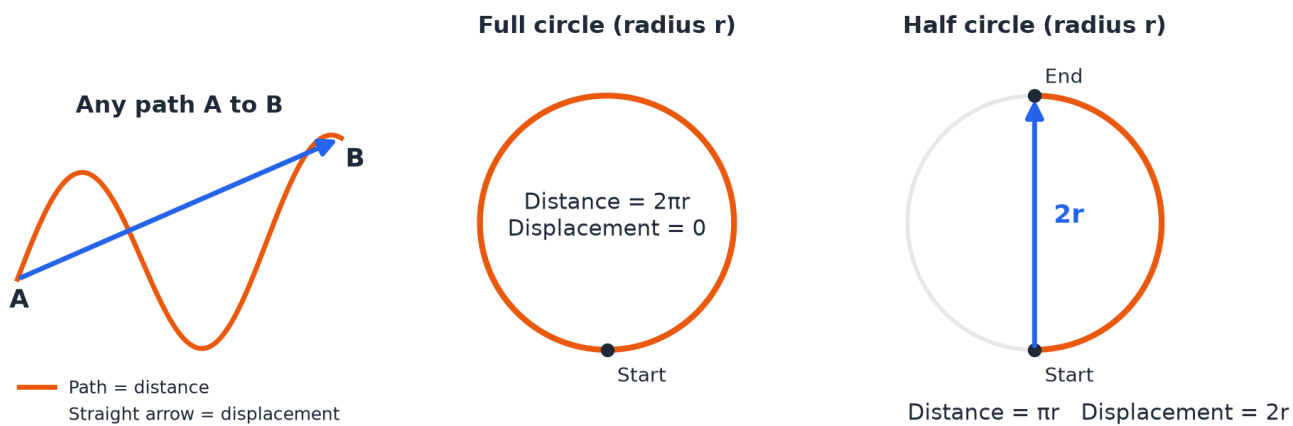
Distance = 3 + 4 = 7 m Displacement = 5 m



- **Distance** = the total length of the path you actually walked = 3 + 4 = **7 m**. It has no direction (scalar).

- **Displacement** = the straight-line gap from where you started to where you ended, **with direction** = $\sqrt{3^2 + 4^2}$ = **5 m** towards the shop (vector).

Now walk around a circle and see what happens:



Rule	Why
Displacement $\leq$ distance, always	The straight line is the shortest path. They are equal only for straight-line motion in one direction.
Displacement can be zero; distance can't (once you've moved)	Come back to your starting point and the displacement is 0.
Full circle: distance $2\pi r$, displacement 0	Half circle: distance πr , displacement $2r$ (the diameter).

Worked example — running track

An athlete runs 1 full lap of a 400 m circular track. Distance = **400 m**. Displacement = **0** (back at the start).

Circle of radius 7 m, half a round ($\pi = 22/7$): distance = $\pi r = 22$ m; displacement = $2r =$ **14 m**.

Quick check 10 — say the answers out loud before moving on (answers on the last page)

- A runner completes 2 full laps of a 400 m track. What are the distance and the displacement?
- Can displacement ever be greater than distance?

Lesson 11 · Speed, velocity and the average-speed trap (about 5 min)

- Speed** = distance $\div$ time (scalar): how fast you're going.
- Velocity** = displacement $\div$ time (vector): how fast **and in which direction**.

In the walk above, if it took you 7 s: speed = $7 \text{ m} \div 7 \text{ s} = 1 \text{ m/s}$, but velocity = $5 \text{ m} \div 7 \text{ s} \approx 0.71 \text{ m/s}$ towards the shop.

- Uniform motion**: equal distances in equal time intervals (constant speed).
- Non-uniform motion**: unequal distances in equal time intervals (city traffic).
- A vehicle's **speedometer** shows the speed at that instant; the **odometer** shows the total distance.

Average speed = total distance $\div$ total time — it is **not** the average of the two speeds. Here's why:

Worked example — why 48, not 50

You drive 60 km to a town at 60 km/h, and return at 40 km/h.

Time going = $60/60 = 1$ h. Time returning = $60/40 = 1.5$ h. Total distance = 120 km, total time = 2.5 h.

Average speed = $120 \div 2.5 = 48$ km/h. You spend **more time** at the slower speed, so it drags the average down.

Shortcut for equal distances at speeds x and y : $2xy \div (x + y) = 2 \times 60 \times 40 \div 100 = 48$ ✓

Average **velocity** for the round trip = 0, because the total displacement is 0.

Converting km/h to m/s: $1 \text{ km/h} = 1000 \text{ m} \div 3600 \text{ s} = 5/18 \text{ m/s}$. So $\text{km/h} \times 5/18 = \text{m/s}$, and $\text{m/s} \times 18/5 = \text{km/h}$.

km/h	18	36	54	72	90	108
m/s	5	10	15	20	25	30

Quick check 11 — say the answers out loud before moving on (answers on the last page)

a) What is 90 km/h in m/s? b) A bus goes from A to B at 30 km/h and returns at 60 km/h. What is its average speed?

Lesson 12 · Acceleration (about 4 min)

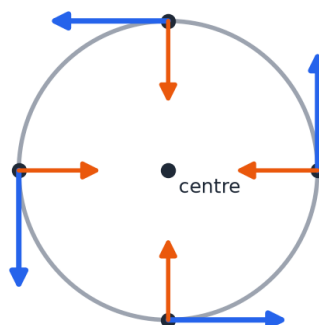
Acceleration = how quickly the velocity changes = (change in velocity) $\div$ time. Watch a car's speedometer as it starts from rest:

Time (s)	0	1	2	3	4	5
Velocity (m/s)	0	4	8	12	16	20

Every second the velocity goes up by 4 m/s, so the acceleration = **4 m/s per second = 4 m/s²**. That is all 'm/s²' means: **m/s gained every second**.

- If the velocity **decreases** (braking), the acceleration is negative. This is called **retardation** or deceleration.
- Velocity changes if the **speed changes OR the direction changes**. So even at constant speed, turning means accelerating.

→ blue: velocity (along the tangent)
same length, direction keeps changing



→ orange: acceleration, always
towards the centre (centripetal)

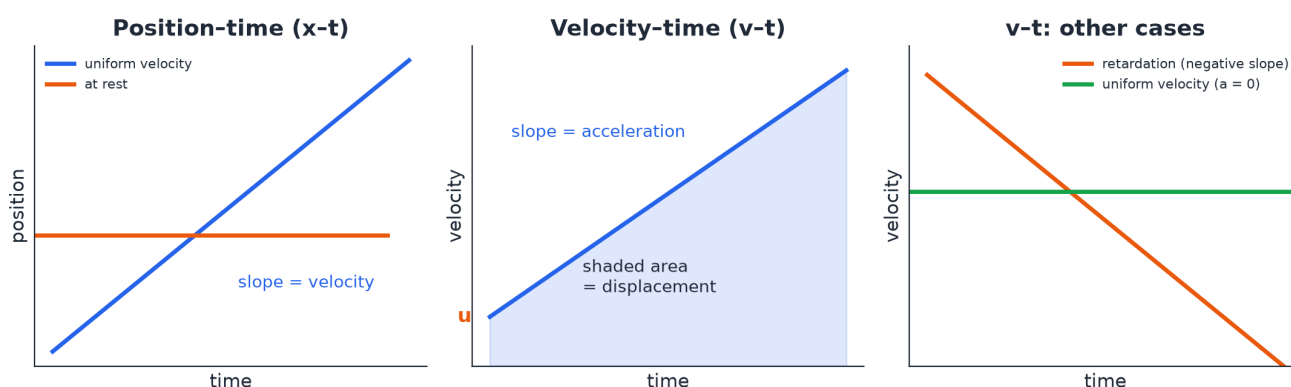
In **uniform circular motion**, the speed stays the same but the direction of the velocity changes at every point. So it **is accelerated motion**. The acceleration always points **towards the centre** and is called **centripetal acceleration**.

Quick check 12 — say the answers out loud before moving on (answers on the last page)

a) Velocity goes from 10 m/s to 30 m/s in 4 s. What is the acceleration? b) Is a body moving in a circle at constant speed accelerating?

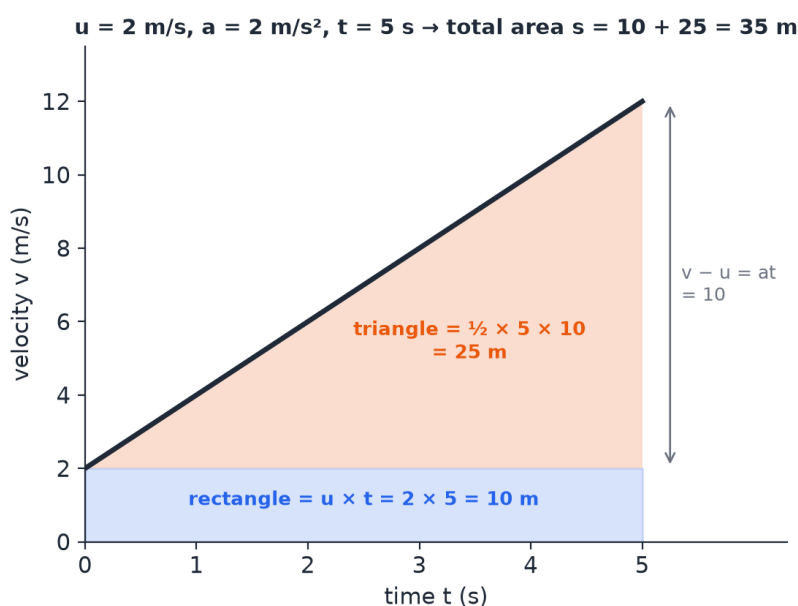
Lesson 13 · Reading motion graphs (about 4 min)

A graph is just the table above drawn as a picture. Only two ideas are needed: **slope** (how steep the line is = rise ÷ run) and **area** (the space under the line).



- On an **x-t graph**, the slope is the velocity: the steeper the line, the faster the motion. A flat line means the object isn't moving.
- On a **v-t graph**, the slope is the acceleration, and the **area under the line is the displacement**. A flat line means constant velocity.

Why is area = displacement? At constant speed, distance = speed × time — the area of a rectangle. When the speed changes, break the area into simple shapes:



Graph	Slope gives	Area under it gives	Horizontal line means
Position-time (x-t)	Velocity	—	Body at rest
Velocity-time (v-t)	Acceleration	Displacement	Constant velocity ($a = 0$)

Graph	Slope gives	Area under it gives	Horizontal line means
Acceleration-time (a-t)	—	Change in velocity	Constant acceleration

Quick check 13 — say the answers out loud before moving on (answers on the last page)

a) What does the slope of an x-t graph give? b) What does a horizontal line on a v-t graph mean?

Lesson 14 · Equations of motion (about 6 min)

There are only 5 quantities in motion problems: **u** (starting velocity), **v** (final velocity), **a** (acceleration), **t** (time), **s** (displacement). All three equations come from the v-t graph you just saw:

#	Equation	Where it comes from	Missing variable
1	$v = u + at$	Slope = $a = (v - u)/t$, rearranged	s
2	$s = ut + \frac{1}{2}at^2$	Area = rectangle ($u \times t$) + triangle ($\frac{1}{2} \times t \times at$)	v
3	$v^2 = u^2 + 2as$	Combine 1 and 2 to remove t	t

3-step recipe for any question: (1) Write down u, v, a, t, s and fill in what's given. (2) Circle the one you need. (3) Pick the equation that **doesn't contain** the quantity you neither know nor need. Code words: 'starts from rest' → $u = 0$. 'Comes to rest' / 'stops' → $v = 0$. 'Dropped' → $u = 0$, $a = g$.

Worked example — A — speeding up

A car starts from rest and reaches 20 m/s in 5 s. Find a and s.

Given: $u = 0$, $v = 20$, $t = 5$. $a = (v - u)/t = 20/5 = \mathbf{4 \text{ m/s}^2}$.

$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 4 \times 25 = \mathbf{50 \text{ m}}$. Check with eq. 3: $20^2 = 0 + 2 \times 4 \times s \rightarrow s = 400/8 = 50$ ✓

Worked example — B — braking

A car at 72 km/h brakes and stops in 4 s. Find the retardation and the stopping distance.

Convert first: $72 \times 5/18 = 20 \text{ m/s}$. Given: $u = 20$, $v = 0$, $t = 4$.

$a = (0 - 20)/4 = \mathbf{-5 \text{ m/s}^2}$ (a retardation of 5 m/s^2).

$s = 20 \times 4 + \frac{1}{2} \times (-5) \times 16 = 80 - 40 = \mathbf{40 \text{ m}}$. Check: $0 = 400 + 2(-5)s \rightarrow s = 40$ ✓

Worked example — C — distance in the nth second

Formula: $s_n = u + a(2n - 1)/2$. A body starts from rest with $a = 2 \text{ m/s}^2$. How far does it go in the 5th second?

$s_5 = 0 + 2 \times (2 \times 5 - 1)/2 = \mathbf{9 \text{ m}}$. Check: distance after 5 s = $\frac{1}{2} \times 2 \times 25 = 25 \text{ m}$; after 4 s = 16 m; $25 - 16 = 9$ ✓

Quick check 14 — say the answers out loud before moving on (answers on the last page)

A train moving at 36 km/h brakes and stops in 5 s. Find (a) its retardation and (b) its stopping distance.

Lesson 15 · Free fall and throwing upward (about 5 min)

Earth pulls every object down with the same acceleration, $g \approx 9.8 \text{ m/s}^2$ (use 10 for quick sums). A feather falls more slowly than a coin only because of **air resistance**. In a **vacuum**, both fall together — Galileo's idea, famously shown on the Moon by astronaut David Scott in 1971 with a hammer and a feather.

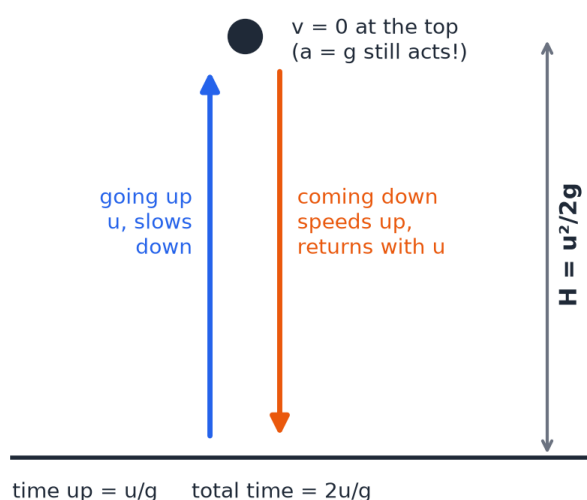
Dropping a stone ($u = 0$, $g = 10$):

Time (s)	1	2	3	4
Speed (m/s) = gt	10	20	30	40
Total fallen (m) = $\frac{1}{2}gt^2$	5	20	45	80
Fallen during that second (m)	5	15	25	35

The distances fallen in each second go 5, 15, 25, 35 — in the ratio **1 : 3 : 5 : 7** (Galileo's law of odd numbers).

Throwing a ball upward at 20 m/s ($g = 10$): gravity slows it by 10 m/s every second.

Time (s)	0	1	2 (top)	3	4
Velocity	20 up	10 up	0	10 down	20 down



- Time up = time down. The ball returns to your hand at the **same speed** it was thrown.
- At the top, the **velocity is 0 but the acceleration is still g** , pointing down. (Very common trap: acceleration is NOT zero at the top.)
- Formulas (from eq. 3 with $v = 0$): **max height $H = u^2/2g$ · time to the top = u/g · total time = $2u/g$.**

Worked example — thrown up

Ball thrown up at 20 m/s: $H = 400/20 = \mathbf{20 \text{ m}}$, time up = **2 s**, total time in the air = **4 s**.

Quick check 15 — say the answers out loud before moving on (answers on the last page)

a) A stone is dropped and falls for 3 s ($g = 10$). What are its speed and the distance fallen? b) A ball is thrown up at 40 m/s. What are its maximum height and total time in the air?

1-page summary — revise from this tomorrow, not from the lessons

Topic	Must remember
SI base units	m, kg, s, A, K, mol, cd (candela = luminous intensity). SI adopted in 1960.
Named units	$N = \text{kg} \cdot \text{m/s}^2$ · $J = N \cdot \text{m}$ · $W = J/s$ · $\text{Pa} = N/\text{m}^2$ · $\text{Hz} = 1/s$ · diopetre = $1/\text{m}$
Big/small units	$\text{\AA} 10^{-10} \text{ m}$ · fermi 10^{-15} m · AU < light year ($9.46 \times 10^{15} \text{ m}$) < parsec (3.26 ly)
Energy & power	1 kWh = $3.6 \times 10^6 \text{ J}$ (energy) · 1 HP = 746 W · 1 cal $\approx 4.186 \text{ J}$ · 1 eV = $1.6 \times 10^{-19} \text{ J}$
Odd ones	Knot = speed · fathom = depth · nautical mile = 1852 m · 1 atm = $1.013 \times 10^5 \text{ Pa}$ = 76 cm Hg
Temperature	$C/5 = (F - 32)/9 = (K - 273)/5$ · $-40^\circ\text{C} = -40^\circ\text{F}$ · body $37^\circ\text{C} = 98.6^\circ\text{F} = 310 \text{ K}$ · a change of $1^\circ\text{C} = 1.8^\circ\text{F}$
Instruments	Hydrometer = liquid density · hygrometer = humidity · ammeter in series, voltmeter in parallel · vernier 0.1 mm · screw gauge 0.01 mm
Scalar / vector	Current and pressure are scalars · weight, momentum and torque are vectors · resultant ranges from $A - B$ to $A + B$
Dimensions	Force MLT^{-2} · work ML^2T^{-2} · power ML^2T^{-3} · pressure $\text{ML}^{-1}\text{T}^{-2}$ · same: work-torque, pressure-stress, impulse-momentum
Distance/displacement	Displacement $\leq$ distance · full circle: displacement 0 · half circle: πr vs $2r$
Speed	Average speed for equal distances = $2xy/(x + y)$ · $\text{km/h} \times 5/18 = \text{m/s}$
Acceleration	Uniform circular motion is accelerated (centripetal, towards the centre)
Graphs	x-t slope = velocity · v-t slope = acceleration · v-t area = displacement
Equations	$v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$ · $s_n = u + a(2n - 1)/2$
Vertical motion	$H = u^2/2g$ · time up = u/g · total = $2u/g$ · at the top $v = 0$ but $a = g$ · free-fall distances per second 1 : 3 : 5

Practice set — 20 questions · target 6 minutes · no looking back

#	Question
1	SI unit of luminous intensity is: (a) lumen (b) lux (c) candela (d) watt
2	Light year is a unit of: (a) time (b) distance (c) speed (d) intensity of light
3	1 kWh equals: (a) 3.6×10^3 J (b) 3.6×10^5 J (c) 3.6×10^6 J (d) 36×10^6 J
4	Which is a vector? (a) speed (b) work (c) momentum (d) energy
5	The relative density of a liquid is measured by a: (a) hygrometer (b) hydrometer (c) lactometer (d) manometer
6	At what temperature are the Celsius and Fahrenheit readings equal? (a) 0 (b) -40 (c) 40 (d) -273
7	A body completes one full circle of radius 7 m. Its displacement is: (a) 44 m (b) 14 m (c) 0 (d) 22 m
8	A body goes half way round a circle of radius 7 m ($\pi = 22/7$). Its distance and displacement are: (a) 22 m, 14 m (b) 14 m, 22 m (c) 44 m, 0 (d) 22 m, 0
9	The slope of a velocity-time graph gives: (a) displacement (b) speed (c) acceleration (d) jerk
10	The area under a velocity-time graph gives: (a) acceleration (b) displacement (c) force (d) momentum
11	A car accelerates from rest at 2 m/s^2 for 10 s. The distance covered is: (a) 20 m (b) 50 m (c) 100 m (d) 200 m
12	54 km/h in m/s is: (a) 12 (b) 15 (c) 18 (d) 20
13	A ball is thrown up at 30 m/s ($g = 10 \text{ m/s}^2$). Its maximum height is: (a) 30 m (b) 45 m (c) 90 m (d) 60 m
14	Which quantity has the same dimensions as work? (a) power (b) force (c) torque (d) momentum
15	A person goes at 60 km/h and returns the same distance at 40 km/h. Their average speed is: (a) 50 (b) 48 (c) 45 (d) 52 km/h
16	Pascal is the unit of: (a) force (b) pressure (c) energy (d) power
17	The least count of a screw gauge is: (a) 0.1 mm (b) 0.01 mm (c) 0.1 cm (d) 1 mm
18	Which is NOT a unit of distance? (a) parsec (b) AU (c) knot (d) light year
19	The SI unit of the power of a lens is: (a) watt (b) dioptre (c) metre (d) joule
20	In uniform circular motion, which stays constant? (a) velocity (b) acceleration vector (c) speed (d) displacement

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Practice set

#	Ans	Why (and which lesson to re-read if wrong)
1	c	Candela is the base unit; lumen = flux, lux = illuminance (Lesson 2)
2	b	Distance light travels in a year (Lesson 4)
3	c	$1000 \text{ W} \times 3600 \text{ s}$ (Lesson 4)
4	c	Momentum = mass $\times$ velocity has a direction (Lesson 7)
5	b	Hygrometer = humidity, lactometer = milk (Lesson 6)
6	b	$x/5 = (x - 32)/9 \rightarrow x = -40$ (Lesson 5)
7	c	Back at the start (Lesson 10)
8	a	$\pi r = 22 \text{ m}$, $2r = 14 \text{ m}$ (Lesson 10)
9	c	Lesson 13
10	b	Lesson 13
11	c	$s = \frac{1}{2} \times 2 \times 100 = 100 \text{ m}$ (Lesson 14)
12	b	$54 \times 5/18 = 15$ (Lesson 11)
13	b	$u^2/2g = 900/20 = 45 \text{ m}$ (Lesson 15)
14	c	Both are force $\times$ distance = ML^2T^{-2} (Lesson 8)
15	b	$2 \times 60 \times 40/100 = 48$ (Lesson 11)
16	b	N/m^2 (Lesson 3)
17	b	Vernier is 0.1 mm (Lesson 6)
18	c	Knot is a speed (Lesson 4)
19	b	1/focal length in m (Lesson 3)
20	c	The direction changes, the speed doesn't (Lesson 12)

Quick checks

QC	Answers
1	a) a number and a unit b) it differs from person to person, so it isn't a fixed standard c) International System of Units
2	a) derived (m/s) b) m^3 c) kelvin (K)
3	a) $600 \div 120 \text{ s} = 5 \text{ W}$ b) 100 N on 0.5 m^2 (200 Pa vs 50 Pa) c) Pa
4	a) $2 \times 3 = 6$ units (kWh) b) $\text{AU} < \text{light year} < \text{parsec}$ c) speed
5	a) $50/5 = 10 \rightarrow \text{F} - 32 = 90 \rightarrow 122 \text{ }^\circ\text{F}$ b) $300 - 273 = 27 \text{ }^\circ\text{C}$ c) $20 \times 1.8 = 36 \text{ }^\circ\text{F}$
6	a) hygrometer b) in parallel (across the device) c) screw gauge (0.01 mm vs 0.1 mm)
7	a) $\sqrt{36 + 64} = 10 \text{ N}$ b) from 2 N to 14 N c) no — current is a scalar
8	a) $[\text{M L T}^{-1}]$ b) stress c) no — it's a ratio of two speeds, so it's dimensionless
10	a) distance 800 m, displacement 0 b) never
11	a) $90 \times 5/18 = 25 \text{ m/s}$ b) $2 \times 30 \times 60/90 = 40 \text{ km/h}$
12	a) $(30 - 10)/4 = 5 \text{ m/s}^2$ b) yes — its direction keeps changing
13	a) velocity b) constant velocity (zero acceleration)
14	$u = 10 \text{ m/s}$, $v = 0$, $t = 5 \rightarrow a = -2 \text{ m/s}^2$ (retardation 2 m/s^2); $s = 10 \times 5 - \frac{1}{2} \times 2 \times 25 = 25 \text{ m}$
15	a) $v = 30 \text{ m/s}$, $h = \frac{1}{2} \times 10 \times 9 = 45 \text{ m}$ b) $H = 1600/20 = 80 \text{ m}$; total time = $2 \times 40/10 = 8 \text{ s}$

After you finish

Send me your practice-set score, your time, and the numbers you got wrong. I'll tell you which lessons to revisit and give you 5 targeted questions.